

Lebesgue Integration & Measure-Theoretic Probability for Econometricians

Eliab Luvanda

2025-08-09

Table of contents

1	Motivation: Why Measure Theory in Econometrics?	3
2	Measure Theory Basics	3
2.1	Sets & Sigma-Algebras	3
2.2	Measures	3
3	Lebesgue Integration	3
3.1	Motivation	3
3.2	Definition	3
3.3	Probabilistic Interpretation	4
4	Measure-Theoretic Probability	4
4.1	Probability Space	4
4.2	Random Variables	4
4.3	Distribution	4
5	Practical Econometric Applications	5
6	Key Theorems for Econometricians	5
6.1	Dominated Convergence Theorem (DCT)	5
6.2	Fubini's Theorem	5
7	Essential Reading	5
8	Concluding Remarks	6
9	Practice Problems	6

10 Visualizations	6
10.1 1. Venn Diagram (Classical/Frequentist Events)	6
10.2 Relative Frequency Plot (Law of Large Numbers)	8
10.3 Bayesian Updating (Prior and Posterior PDFs)	9
10.4 Probability Space Diagram (Measure-Theoretic)	9
10.5 Probability of Hitting a Subset: Measure-Theoretic Perspective	11

List of Figures

1 Venn Diagram of Events A and B	7
2 Empirical Probability Converges to True Probability	8
3 Bayesian Updating: Prior, Likelihood, and Posterior Distributions	9
4 Dartboard Visualization: Sample Space Ω and Event $A = \{x < 0\}$	10
5 Dartboard Visualization of computing a probability measure of an event A	12

List of Tables

1 Summary Table: Riemann vs Lebesgue Integration	5
--	---

1 Motivation: Why Measure Theory in Econometrics?

- **Econometric models** require integration over probability spaces, especially for expectations and variances.
- Classical Riemann integration is limited; Lebesgue integration generalizes the concept, handling more complex random variables and probability spaces.
- Measure-theoretic probability provides a rigorous framework for advanced econometric theory.

2 Measure Theory Basics

2.1 Sets & Sigma-Algebras

- **Sample Space (Ω):** Universe of outcomes.
- **Sigma-Algebra ($\mathcal{F}$):** A collection of subsets of Ω (including Ω and $\emptyset$) closed under countable unions and complements. Ensures “measurability” of events.

2.2 Measures

- **Measure μ :** Function $\mu : \mathcal{F} \rightarrow [0, \infty]$, assigning “size” to sets.
- **Probability Measure P :** Special case, $P(\Omega) = 1$, representing certainty.

3 Lebesgue Integration

3.1 Motivation

- Riemann integral partitions the domain (horizontal slices).
- Lebesgue integral partitions the range (vertical slices); better suited for complex/discontinuous functions.

3.2 Definition

3.2.1 Simple Function

A simple function $s(x)$ is:

$$s(x) = \sum_{k=1}^n a_k \mathbb{I}_{A_k}(x)$$

where $A_k \in \mathcal{F}$, $a_k \geq 0$.

3.2.2 Lebesgue Integral (non-negative f)

$$\int_{\Omega} f d\mu = \sup \left\{ \int_{\Omega} s d\mu : 0 \leq s \leq f, s \text{ is simple} \right\}$$

3.2.3 Key Properties

- **Linearity:** $\int (f + g) d\mu = \int f d\mu + \int g d\mu$
- **Monotonicity:** If $f \leq g$, then $\int f d\mu \leq \int g d\mu$
- **Dominated Convergence Theorem (DCT)**
- **Fatou's Lemma**
- **Fubini's Theorem**

3.3 Probabilistic Interpretation

- **Expectation:**

$$E[X] = \int_{\Omega} X(\omega) P(d\omega)$$

Applies to discrete, continuous, and mixed distributions.

4 Measure-Theoretic Probability

4.1 Probability Space

- **Triple $(\Omega, \mathcal{F}, P)$:**
 - Ω : Sample space
 - $\mathcal{F}$: Sigma-algebra
 - P : Probability measure

4.2 Random Variables

- **Random Variable (X):** Measurable function $X : \Omega \rightarrow \mathbb{R}$
- **Measurability:** For all Borel sets B , $X^{-1}(B) \in \mathcal{F}$

4.3 Distribution

- **Pushforward Measure μ_X of X :**

$$\mu_X(B) = P(X \in B) = P(X^{-1}(B))$$

5 Practical Econometric Applications

- **Expectations, variances, covariances:** All defined as Lebesgue integrals.
- **Laws of Large Numbers and Central Limit Theorem:** Measure-theoretic convergence (almost sure, in probability) is key.
- **Likelihood estimation:** Supports both continuous and mixed models.

6 Key Theorems for Econometricians

6.1 Dominated Convergence Theorem (DCT)

If $f_n \rightarrow f$ and $|f_n| \leq g$ (integrable),

$$\lim_{n \rightarrow \infty} \int f_n d\mu = \int f d\mu$$

6.2 Fubini's Theorem

Interchange order of integration:

$$\int_{\Omega} \int_{\Theta} f(x, y) d\nu(y) d\mu(x) = \int_{\Theta} \int_{\Omega} f(x, y) d\mu(x) d\nu(y)$$

Table 1: Summary Table: Riemann vs Lebesgue Integration

Feature	Riemann Integral	Lebesgue Integral
Partitioning	Domain (subintervals)	Range (value slices)
Functions handled	Continuous	Measurable
Probabilistic Models	Limited	Natural & robust
Convergence Theorems	Few	DCT, Fatou's, etc.
Use for Expectations	Simple RVs only	General probability models

7 Essential Reading

- Billingsley, P. *Probability and Measure*
- Dudley, R. *Real Analysis and Probability*
- Chung, K.L. *A Course in Probability Theory*

8 Concluding Remarks

- **Lebesgue integration** and **measure-theoretic probability** are foundational for econometric theory, supporting advanced models, convergence concepts, expectations, and asymptotics.

9 Practice Problems

1. **Probability Spaces:** Construct a probability space for a Markov process, identify measurable events.
 2. **Expectation Calculation:** If $X = \mathbb{I}_A + 2\mathbb{I}_B$, $A, B \in \mathcal{F}$, compute $E[X]$.
 3. **Dominated Convergence:** State and apply the Dominated Convergence Theorem to a sequence of expected values.
-

End of Notes

10 Visualizations

10.1 1. Venn Diagram (Classical/Frequentist Events)

The downloaded binary packages are in
`/var/folders/hw/_50mvf05767952_56w981ssc0000gn/T//RtmpmhU2JS/downloaded_packages`

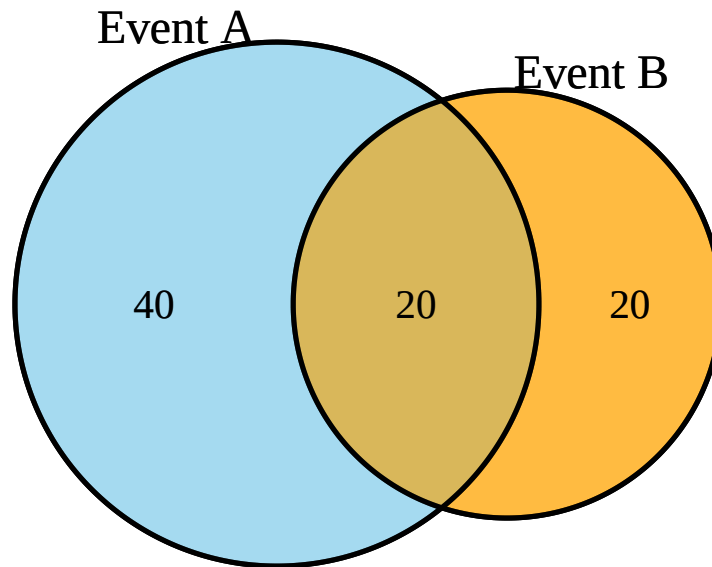


Figure 1: Venn Diagram of Events A and B

Purpose:

A Venn diagram visually demonstrates how events and their relationships (intersection, union, disjointness) are handled in classical and frequentist probability.

Detailed Explanation:

- **Events:** Each circle represents a possible event (e.g., getting a head in a coin toss, or an ace in a card draw). - **Intersection:** The overlapping region shows outcomes shared by both events (e.g., outcomes where both “A” and “B” occur). - **Union:** The total area covered by circles represents “A or B.” - **Disjointness:** If circles don’t overlap, events are mutually exclusive. - Use this plot to discuss how probabilities of unions and intersections are calculated (e.g., $P(A \cup B) = P(A) + P(B) - P(A \cap B)$).

10.2 Relative Frequency Plot (Law of Large Numbers)

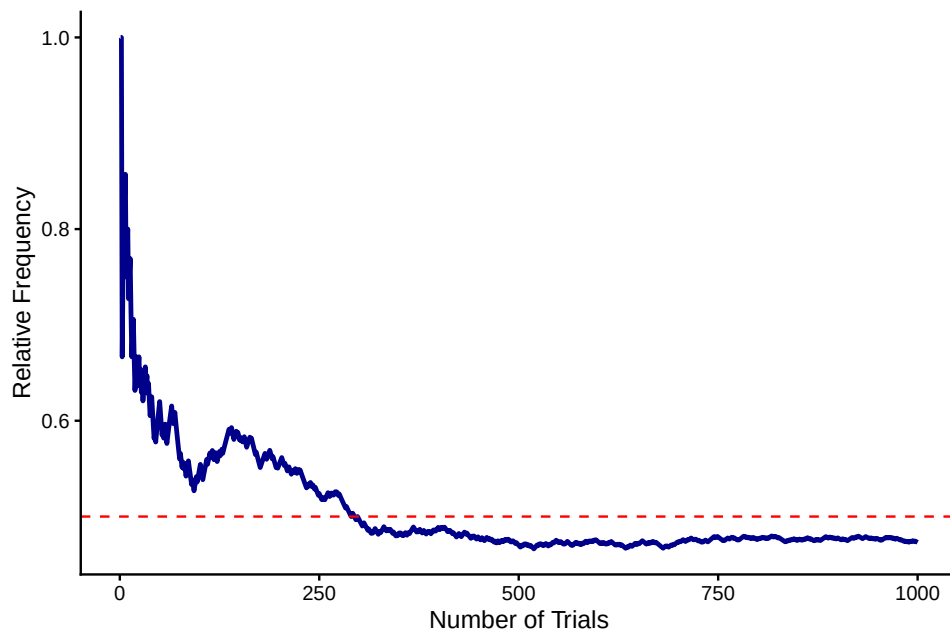


Figure 2: Empirical Probability Converges to True Probability

Purpose:

Shows how empirical (experimentally observed) frequencies converge to theoretical probabilities as the number of trials increases—a foundation of frequentist probability and statistics.

Detailed Explanation:

- **Empirical Line:** The blue line tracks the proportion of heads in repeated coin tosses.
- **Theoretical Probability:** The dashed red line marks the true value ($p = 0.5$ for a fair coin).
- **Convergence:** As the number of trials increases, the empirical proportion stabilizes around the theoretical probability, visually illustrating the Law of Large Numbers.

10.3 Bayesian Updating (Prior and Posterior PDFs)

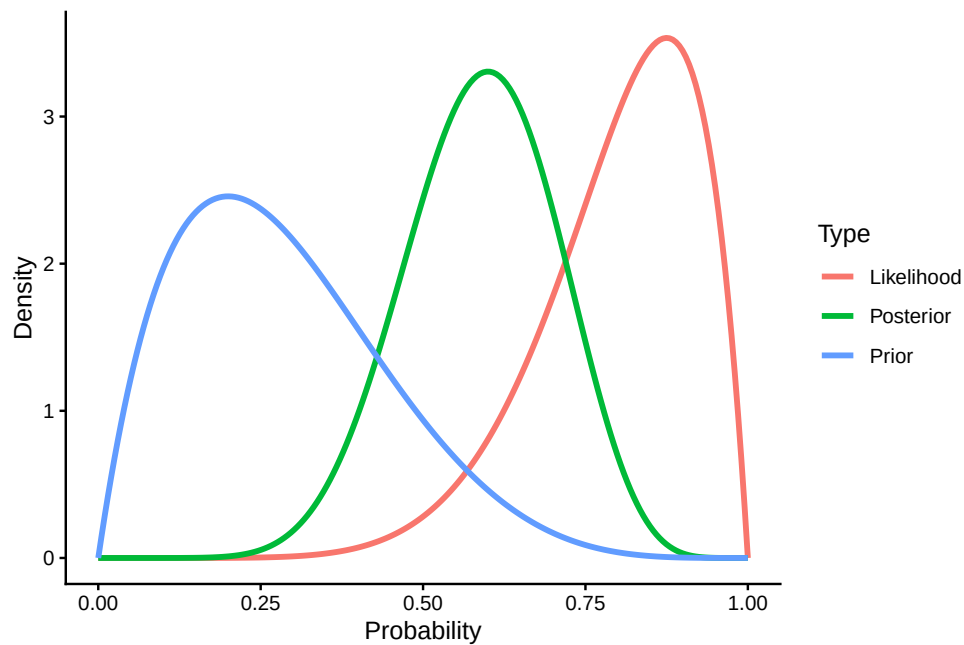


Figure 3: Bayesian Updating: Prior, Likelihood, and Posterior Distributions

Purpose: Visualizes how Bayesian updating modifies beliefs (prior) based on new evidence (likelihood) to produce a posterior distribution.

Detailed Explanation:

- **Prior Distribution:** Represents initial beliefs about a parameter (e.g., probability of success).
- **Likelihood:** Represents the evidence or data observed (e.g., number of successes in trials).
- **Posterior Distribution:** Updated beliefs after considering the likelihood, combining prior and evidence.
- **Interpretation:** The posterior reflects how the prior is adjusted by the likelihood, showing the Bayesian approach to probability and inference.

10.4 Probability Space Diagram (Measure-Theoretic)

Measure-Theoretic Probability Space can handle infinite & uncountable sample spaces, such as a dart board.

Let's consider a **dart board** as a classic example of an infinite and uncountable probability space, which is a fundamental concept in measure-theoretic probability.

10.4.1 Description

- **Sample Space (Ω)**: The set of all points on the dart board, mathematically represented as the set of all ordered pairs

$$(x, y)$$

in the unit disk:

$$\Omega = \{(x, y) \mid x^2 + y^2 \leq 1\}$$

This set is **uncountably infinite**, meaning there are infinitely many points, and you cannot list them in a sequence.

- **Events ($A \subset \Omega$)**: An event is any measurable subset of the dart board — for instance, a specific region, such as a sector or an annulus.
- **Probability Measure ($\mathbb{P}$)**: The probability of “hitting” a subset A is defined by the area of A divided by the total area of the dart board. Formally,

$$\mathbb{P}(A) = \frac{\text{Area}(A)}{\pi}$$

(assuming the dart lands uniformly at random on the board of area π .)

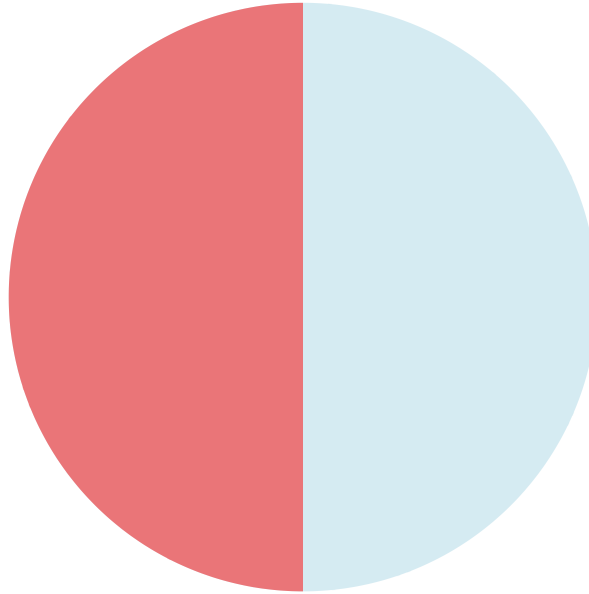


Figure 4: Dartboard Visualization: Sample Space Ω and Event $A = \{x < 0\}$

10.4.2 Key Ideas from Probability Theory:

- **Sample Space (Ω):** The unit disk shows every possible outcome—the set of all points where a dart can land. This is an infinite, uncountable sample space.
- **Event ($A = (x, y) : x < 0$):** The left half is highlighted in red—representing a measurable event (a subset of possible outcomes).
- **Probability Measure:** In continuous spaces, probability is assigned by area:

For this symmetric case, the probability of hitting the left half is 50%. This visualization demonstrates the essence of measure-theoretic probability, where events are regions of a continuous space, and likelihoods are determined by their size relative to the whole sample space.

10.5 Probability of Hitting a Subset: Measure-Theoretic Perspective

In measure-theoretic probability theory, the **sample space** (Ω) can be a continuous, uncountable set of outcomes—such as every possible point where a dart can land on a circular board. Suppose we are interested in the probability that the dart lands inside a **smaller circle** (event A) that lies within the larger circle (the sample space).

This setting formalizes probability in terms of **measures** (such as area), rather than counting discrete outcomes. Specifically:

- **Sample Space (Ω):**
The entire large circle, representing all possible points where the dart may land.
- **Event (A):**
Any measurable subset of Ω . In this scenario, A is the smaller circle (possibly offset from the center) entirely or partially within the larger one.
- **Probability Measure:**
The probability of hitting any point in A is given by the ratio of the area of A to the area of Ω —assuming the dart lands uniformly at random:

$$\mathbb{P}(A) = \frac{\text{Area}(A)}{\text{Area}(\Omega)}$$

If the smaller circle (radius r) is entirely inside the larger circle (radius R):

$$\mathbb{P}(A) = \frac{\pi r^2}{\pi R^2} = \left(\frac{r}{R}\right)^2$$

If the inner circle overlaps the edge of the larger one, $\mathbb{P}(A)$ is the ratio of the overlapping area to the area of Ω .

Interpretation:

This approach encapsulates the essence of measure-theoretic probability:

- The probability of an event in a continuous space is computed by comparing its “size” (area or measure) to that of the full sample space.
- Events need not be centered, nor do they have to be simple shapes—they can be any measurable region within Ω .

Visualization:

The following graph (see next section) depicts this relationship, with the **larger circle** representing all possible outcomes and the **smaller, possibly offset circle** highlighting the subset of interest whose probability we are computing.

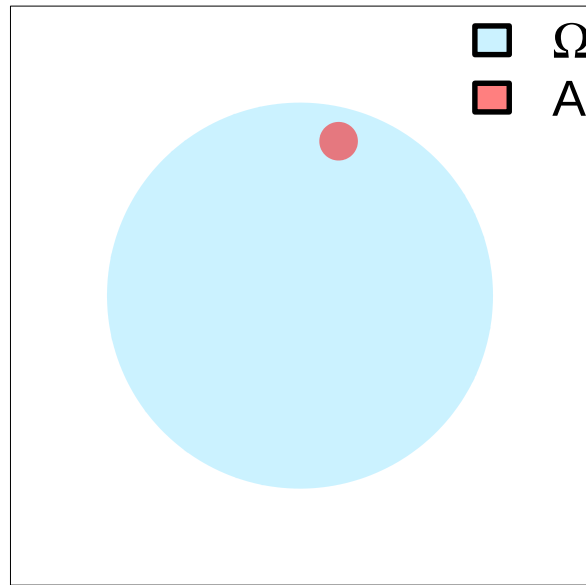


Figure 5: Dartboard Visualization of computing a probability measure of an event A

10.5.1 Key Measure-Theoretic Probability Ideas Illustrated by the Graph

- **Sample Space (Ω):** The larger circle represents all possible outcomes where the dart can land.
- **Event (A):** The smaller circle, possibly offset, represents a specific measurable event within the sample space.
- **Probability Measure:** The shaded area of the smaller circle relative to the larger circle illustrates how we compute the probability of event A by comparing its area to that of Ω .

This visualization captures the essence of measure-theoretic probability, where events are defined in terms of their measurable size within a continuous space.